

MIRAMICHI, NB, Canada

WMO: 717440

Lat: 47.0094N

Lon: 65.4650W

Elev: 111

StdP: 14.64

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.5	-5.7	-19.5	1.9	-8.4	-14.8	2.5	-3.7	27.2	32.2	23.2	27.4	5.4	260	0.617

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	86.2	69.3	82.7	67.6	79.5	65.8	72.1	81.5	70.3	78.5	68.6	76.1	11.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.2	107.7	76.2	67.4	101.3	74.4	65.8	95.6	72.9	36.0	81.7	34.4	78.8	32.9	76.2	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.5	19.4	17.1	DB	-18.2	92.1	4.4	3.2	-21.4	94.5	-24.0	96.3	-26.5	98.1	-29.7	100.5	(4)
(5)			WB	-18.6	76.1	4.4	2.5	-21.8	77.9	-24.3	79.3	-26.8	80.7	-30.0	82.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.2	15.6	17.1	26.5	38.4	50.6	60.8	68.0	66.2	58.2	46.3	34.7	22.8		
	DBStd	19.95	11.29	10.83	9.54	6.98	7.40	7.18	5.30	5.38	6.92	7.28	8.86	9.95		
	HDD50	4635	1066	921	728	354	82	5	0	0	9	162	466	842		
	HDD65	8578	1531	1341	1192	797	451	165	29	49	224	582	910	1307		
	CDD50	1805	0	0	1	6	101	330	558	502	255	46	6	0		
	CDD65	274	0	0	0	0	5	40	122	86	20	1	0	0		
	CDH74	2369	0	0	1	5	103	435	976	687	156	6	0	0		
	CDH80	619	0	0	0	1	21	126	256	178	37	0	0	0		
	WSAvg	7.7	7.7	7.9	8.6	8.3	8.1	7.5	6.9	6.7	7.0	7.6	7.9	7.7		
	PrecAvg	38.0	3.0	2.6	2.9	2.5	3.5	3.1	3.5	3.2	3.2	4.0	3.4	3.2		
(15) Precipitation	PrecMax	48.4	5.3	5.1	6.1	5.6	6.3	5.7	7.5	6.9	8.0	7.1	5.9	8.3		
	PrecMin	26.6	0.5	0.2	0.7	0.9	0.6	1.1	0.4	0.9	0.8	0.8	1.3	0.5		
	PrecStd	4.6	1.3	1.2	1.3	1.2	1.4	1.1	1.7	1.5	1.7	1.7	1.2	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.2	48.5	57.0	69.4	83.1	89.0	90.0	89.9	85.3	73.7	63.8	50.6		
		MCWB	48.7	44.3	46.1	52.5	61.6	69.4	71.3	70.6	69.9	62.4	60.7	49.1		
	2%	DB	41.9	40.7	47.4	60.1	76.3	83.7	86.0	84.6	78.3	66.5	58.2	44.4		
		MCWB	39.8	37.1	40.4	49.3	59.1	66.3	70.1	68.9	66.9	58.9	54.7	42.7		
	5%	DB	36.6	36.3	43.3	55.2	70.2	78.8	82.7	80.9	73.8	62.2	52.7	39.6		
		MCWB	34.5	33.4	37.6	46.0	56.7	63.7	68.2	67.4	63.9	56.0	49.0	37.2		
	10%	DB	32.2	33.2	39.6	50.8	65.1	74.8	79.5	77.6	70.4	58.1	47.5	35.8		
		MCWB	30.6	31.0	34.9	42.8	54.0	61.8	66.3	65.6	62.6	52.8	44.2	34.1		
	0.4%	WB	48.9	45.6	49.5	57.7	66.2	73.4	75.1	74.1	72.2	64.3	61.4	49.4		
		MCDB	50.1	48.3	53.7	65.3	76.9	86.1	85.3	83.6	81.1	69.4	63.6	50.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.3	37.8	42.0	52.0	62.0	69.5	72.4	71.6	69.2	60.5	55.5	43.0		
		MCDB	41.5	40.1	46.1	56.7	70.5	78.6	81.6	80.7	74.9	64.7	57.5	44.3		
	5%	WB	34.5	33.9	38.3	47.3	58.7	67.0	70.6	69.7	66.6	56.8	49.7	37.7		
		MCDB	36.3	35.8	42.1	53.4	67.1	74.9	79.0	77.2	72.0	61.3	52.0	38.9		
	10%	WB	30.7	31.2	35.3	43.6	55.2	64.3	68.7	67.8	63.9	53.7	44.6	34.2		
		MCDB	32.0	33.2	39.0	50.0	64.1	71.4	76.6	74.9	68.8	57.5	47.3	35.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.4	19.3	17.8	17.4	20.3	20.2	19.3	19.5	19.7	16.9	13.8	14.7		
		MCDBR	22.4	19.7	21.7	26.3	30.8	27.7	24.0	24.4	23.8	22.6	20.1	15.7		
		MCWBR	22.1	17.1	16.1	17.2	17.0	14.3	11.4	12.2	14.1	15.8	17.8	14.8		
		MCDBR	22.4	17.9	19.3	23.2	26.3	22.6	20.2	20.0	19.9	19.5	18.9	15.4		
	5% WB	MCWBR	22.4	16.4	15.7	17.5	16.4	13.7	10.9	11.7	13.5	15.5	18.2	15.1		
(40) Clear-Sky Solar Irradiance	taub		0.283	0.281	0.317	0.357	0.369	0.391	0.417	0.392	0.360	0.328	0.300	0.281		
	taud		2.405	2.379	2.390	2.381	2.404	2.338	2.305	2.366	2.443	2.526	2.552	2.487		
	Ebn at Noon		258	284	288	285	284	277	268	270	270	264	252	247		
	Edh at Noon		22	27	32	36	36	39	40	36	30	24	19	18		
(44) All-Sky Solar Radiation	RadAvg		424	671	1057	1347	1605	1730	1752	1532	1204	755	413	311		
	RadStd		44	72	73	123	160	131	124	100	105	68	51	33		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page